PROJECT PROPOSAL

Employee Performance & HR Analytics Dashboard

A Data-Driven Approach to Workforce Insights

Department: Data Analytics
Submitted to: Management / HR Department
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1. Executive Summary

This proposal outlines a comprehensive Employee Performance & HR Analytics project designed to convert raw human resource data into actionable business intelligence. Most organizations collect significant employee data through HRIS, payroll, and attendance systems, but very little of it is converted into decisions that improve productivity, retention, or workforce planning.

The proposed project will build an end-to-end analytics solution using SQL for data extraction, Python for data cleaning and predictive modelling, Excel for ad-hoc analysis, and Power BI for executive dashboards. The final deliverable will be an interactive dashboard combined with a predictive attrition model that helps leadership identify at-risk employees, optimize hiring, and improve overall workforce performance.

Estimated duration: 8 weeks. Estimated team: 2 to 3 analysts. Expected outcome: a reusable analytics platform that pays back its cost within the first quarter of deployment through improved retention and reduced hiring overhead.

2. Problem Statement

Despite collecting employee data across multiple systems, the company currently faces several challenges:

- HR data is scattered across spreadsheets, payroll software, and attendance systems with no single source of truth.
- Manual reporting consumes significant analyst time every month and leaves little room for proactive analysis.
- Leadership lacks visibility into attrition patterns, department-level performance, and compensation fairness.
- There is no early-warning mechanism to flag employees likely to leave the organization.
- Hiring decisions and promotions rely on intuition rather than data-backed evidence.

Without a structured analytics layer, the company continues to make reactive rather than predictive workforce decisions.

3. Project Objectives

1. Consolidate employee data from multiple sources into a single, clean analytical dataset.
2. Build an interactive Power BI dashboard for HR leadership and department heads.
3. Develop a predictive model in Python to estimate attrition risk per employee.
4. Identify key drivers of employee performance, attrition, and engagement.
5. Produce monthly automated reports replacing manual Excel reporting cycles.
6. Enable data-driven decision-making across hiring, promotions, and compensation.

4. Scope of the Project

4.1 In Scope

- Employee demographic, role, and tenure analysis.
- Attendance, leave, and overtime trend analysis.
- Performance review score analysis and benchmarking.
- Attrition analysis and predictive risk modelling.
- Compensation analysis (pay equity, salary bands, market comparison).
- Interactive Power BI dashboard with role-based access.

4.2 Out of Scope

- Recruitment portal integration or applicant tracking system development.
- Payroll processing or tax computation.
- Personally identifiable information will be anonymized; no individual surveillance.

5. Methodology

The project will follow a five-phase data analytics lifecycle:

Phase	Tools	Activities
1. Data Collection	SQL, Python	Extract data from HRIS, payroll DB, attendance logs.
2. Data Cleaning	Python (Pandas)	Handle missing values, standardize formats, anonymize PII.
3. Exploratory Analysis	Excel, Python	Univariate, bivariate, and correlation analysis.
4. Modelling	Python (Scikit-learn)	Logistic regression and Random Forest for attrition prediction.
5. Visualization	Power BI	Interactive dashboards with drill-down capability.

6. Tools and Technology Stack

Tool	Purpose
SQL	Data extraction from HRIS and payroll databases. Writing aggregated views and stored procedures.
Python	Data cleaning (Pandas, NumPy), statistical analysis, and machine learning (Scikit-learn, XGBoost).
Excel	Quick ad-hoc analysis, pivot tables, and scenario modelling for HR business partners.
Power BI	Final dashboard delivery, scheduled refresh, and row-level security for managers.
Jupyter	Documenting analytical notebooks for reproducibility and knowledge transfer.

7. Deliverables

- Cleaned, consolidated employee master dataset (SQL views + CSV extract).
- Python notebooks documenting data cleaning, EDA, and model building.
- Attrition prediction model with documented accuracy and feature importance.
- Interactive Power BI dashboard with department, location, and role filters.
- Monthly automated HR analytics report (PDF + Power BI link).
- Final project documentation and user guide for HR business partners.

8. Project Timeline (8 Weeks)

Week	Activity	Milestone
Week 1	Requirement gathering & stakeholder interviews	Signed-off scope document
Week 2	Data extraction and SQL view creation	Source-to-target mapping
Week 3	Data cleaning and anonymization	Master analytical dataset
Week 4	Exploratory data analysis	EDA report with key findings
Week 5	Predictive model development	Attrition model v1 + accuracy report
Week 6	Power BI dashboard design	Draft dashboard for review
Week 7	User acceptance testing & refinement	Final dashboard
Week 8	Training, handover & documentation	Live deployment + user guide

9. Team & Roles

Role	Responsibility
Project Lead / Senior Analyst	Overall delivery, stakeholder communication, model validation.
Data Analyst (SQL + Python)	Data extraction, cleaning, EDA, and notebook documentation.
BI Developer	Power BI dashboard development, DAX measures, security setup.
HR Business Partner (Stakeholder)	Requirements input, data validation, business interpretation.

10. Expected Outcomes & KPIs

- Reduce monthly HR reporting effort by approximately 70%.
- Identify the top 5 drivers of attrition within the company.
- Achieve 80%+ accuracy on the attrition prediction model.
- Decrease voluntary attrition by 10-15% within two quarters of deployment.
- Provide every department head with a self-service dashboard.
- Improve pay equity visibility across roles and locations.

11. Risks & Mitigation

Risk	Impact	Mitigation
Data quality issues	High	Dedicated cleaning phase + validation checks with HR team.
Privacy / PII exposure	High	Anonymize personal identifiers; role-based access in Power BI.
Low stakeholder adoption	Medium	Early involvement of HR partners + dashboard training sessions.
Model bias	Medium	Audit features for sensitive attributes; document limitations.
Tool licensing delays	Low	Confirm Power BI Pro licenses before Week 6.

12. Estimated Budget

The project relies primarily on existing internal resources and tooling. Indicative budget heads (to be finalized with finance):

Item	Estimate
Analyst time (2 analysts x 8 weeks)	Internal allocation
Power BI Pro licenses	As per existing subscription
Cloud compute (if required for model training)	Minimal, on-demand
Training & change management workshops	1 to 2 sessions

13. Conclusion

Investing in employee analytics is no longer optional. Companies that act on workforce data outperform those that rely on intuition, especially in retention and productivity. This project delivers a focused, low-risk, high-impact analytics solution that the company can operationalize within two months.

With the right combination of SQL, Python, Excel, and Power BI, the organization will move from reactive HR reporting to predictive workforce intelligence, giving leadership the visibility they need to make confident, evidence-based people decisions.